

INVERSE PROBLEMS FOR QUASI-LINEAR ELLIPTIC SYSTEMS MODELING ELECTROLYSERS

Giovanni S. Alberti

University of Genoa

In collaboration with Matteo Santacesaria & Wadim Gerner



Università
di Genova



Funded by
the European Union



European Research Council
Established by the European Commission

MOTIVATION

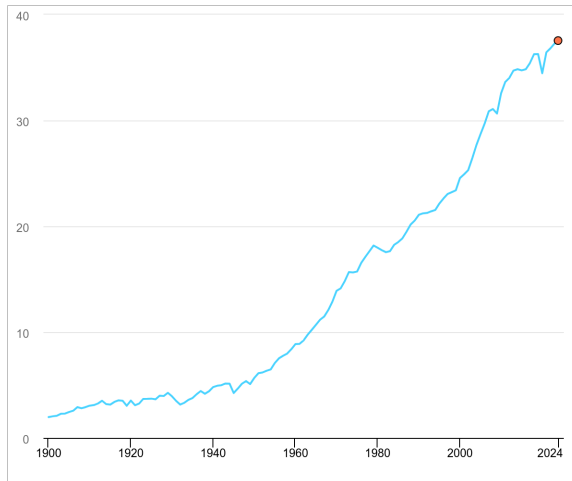
- ① Hydrogen used in many industrial applications (e.g. Ammonia production → fertilisers)
- ② Commonly CO_2 is a by-product in hydrogen production → environmental pollution



FIGURE: Fritz Haber (left) and Carl Bosch (right): Founder's of the Haber-Bosch process for ammonia production, source: Wikimedia

CO₂ POLLUTION

- In 2023, global hydrogen production emitted 920 Mt CO₂ (Global Hydrogen Review 2024)
- In 2023, global energy related CO₂ pollution ~ 37.5 Gt (Global Energy Review 2025)



A POSSIBLE SOLUTION: WATER ELECTROLYSIS

- Cathode: $4\text{H}_2\text{O} + 4\text{e}^- \rightarrow 2\text{H}_2 + 4\text{OH}^-$
- Anode: $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
- Overall: $2\text{H}_2\text{O} \xrightarrow{\text{electricity}} 2\text{H}_2 + \text{O}_2$;
- No CO_2 production!

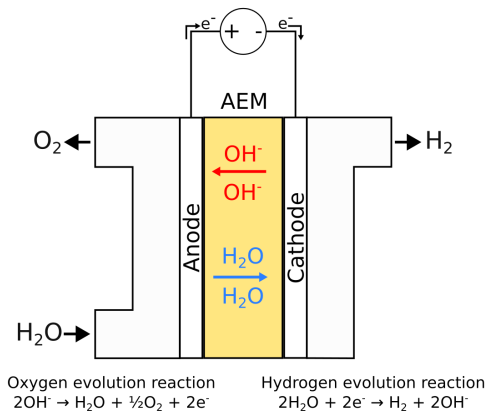


FIGURE: Schematics by Kevin Teenakul

- Electric potential ϕ in electrolyser cell $\leftrightarrow$ lifetime of electrolyser

- Electric potential ϕ in electrolyser cell $\leftrightarrow$ lifetime of electrolyser
- Goal: Describe electric potential ϕ as function of
 - 1 temperature $T = T(x)$
 - 2 ion concentrations $c_i = c_i(x)$, $i = 1, \dots, M$ (M distinct particle types)

- Electric potential ϕ in electrolyser cell $\leftrightarrow$ lifetime of electrolyser
- Goal: Describe electric potential ϕ as function of
 - ① temperature $T = T(x)$
 - ② ion concentrations $c_i = c_i(x)$, $i = 1, \dots, M$ (M distinct particle types)
- $\phi = \phi(T, c, x)$, $c = (c_1, \dots, c_M)$.

THE MODEL EQUATIONS: THE CONCENTRATIONS

- Concentrations determined by elliptic system of PDEs (Fick's law of diffusion)

$$-\operatorname{div}(D_i(T, c, x) \nabla c_i(x)) = g_i(T, c, x)$$

- 1 $D_i(T, c, x)$ = diffusion coefficients (a priori unknown)
- 2 $g_i(T, c, x)$ = source terms $\rightarrow$ some structure may be assumed (Mass action kinetics, Arrhenius law ...)

THE MODEL EQUATIONS: THE CONCENTRATIONS

- Concentrations determined by elliptic system of PDEs (Fick's law of diffusion)

$$-\operatorname{div}(D_i(T, c, x) \nabla c_i(x)) = g_i(T, c, x)$$

- ① $D_i(T, c, x)$ = diffusion coefficients (a priori unknown)
- ② $g_i(T, c, x)$ = source terms $\rightarrow$ some structure may be assumed (Mass action kinetics, Arrhenius law ...)
- ③ Throughout we assume $g_i = 0$ for all i ($g_i \neq 0$ more challenging)

THE MODEL EQUATIONS: THE CONCENTRATIONS

- Concentrations determined by elliptic system of PDEs (Fick's law of diffusion)

$$-\operatorname{div}(D_i(T, c, x) \nabla c_i(x)) = g_i(T, c, x)$$

- ① $D_i(T, c, x)$ = diffusion coefficients (a priori unknown)
- ② $g_i(T, c, x)$ = source terms $\rightarrow$ some structure may be assumed (Mass action kinetics, Arrhenius law ...)
- ③ Throughout we assume $g_i = 0$ for all i ($g_i \neq 0$ more challenging)

$$\operatorname{div}(D_i(T, c, x) \nabla c_i) = 0, \quad i = 1, \dots, M.$$

THE ELECTRIC POTENTIAL

- Potential determined by macroscopic Maxwell's equations

$$\Delta(\phi(T, c, x)) = \rho_f$$

- ρ_f = free-charges

- Potential determined by macroscopic Maxwell's equations

$$\Delta(\phi(T, c, x)) = \rho_f$$

- ρ_f = free-charges = ion-charges

$$\rho_f = \sum_{i=1}^M q_i c_i = q \cdot c, \quad q_i = \text{charge of } i\text{-th particle type (known)}$$

THE ELECTRIC POTENTIAL

- Potential determined by macroscopic Maxwell's equations

$$\Delta(\phi(T, c, x)) = \rho_f$$

- ρ_f = free-charges = ion-charges

$$\rho_f = \sum_{i=1}^M q_i c_i = q \cdot c, \quad q_i = \text{charge of } i\text{-th particle type (known)}$$

- Overall

$$\Delta(\phi(T, c, x)) = q \cdot c$$

THE FORWARD PROBLEM: EXISTENCE

Throughout $\Omega \subset \mathbb{R}^3$ is a bounded $C^{1,\beta}$ -domain and $M \in \mathbb{N}$ a positive integer.

THE FORWARD PROBLEM: EXISTENCE

Throughout $\Omega \subset \mathbb{R}^3$ is a bounded $C^{1,\beta}$ -domain and $M \in \mathbb{N}$ a positive integer.

EXISTENCE

Let $\phi \in C^1$ be globally Lipschitz continuous, $D_i \in C^0 \cap L^\infty$, $1 \leq i \leq M$, $q \in \mathbb{R}^M$. Assume the following two conditions

- ① $0 < D_* \leq D_i$ for some $D_* \in (0, \infty)$ and all $1 \leq i \leq M$,
- ② $0 < \phi_* \leq (\partial_T \phi)(T, c, x)$ for some $\phi_* \in (0, \infty)$ and all T, c, x .

Then for every $(\gamma, \tau) \in \left(W^{\frac{1}{2},2}(\partial\Omega)\right)^{M+1}$ there is a weak solution $(c, T) \in (H^1(\Omega))^{M+1}$ of the system of equations

$$\operatorname{div}(D_i(T, c, x) \nabla c_i) = 0, \quad \Delta(\phi(T, c, x)) = q \cdot c \text{ in } \Omega, \quad (c, T)|_{\partial\Omega} = (\gamma, \tau).$$

THE FORWARD PROBLEM: EXISTENCE

Throughout $\Omega \subset \mathbb{R}^3$ is a bounded $C^{1,\beta}$ -domain and $M \in \mathbb{N}$ a positive integer.

EXISTENCE

Let $\phi \in C^1$ be globally Lipschitz continuous, $D_i \in C^0 \cap L^\infty$, $1 \leq i \leq M$, $q \in \mathbb{R}^M$. Assume the following two conditions

- ① $0 < D_* \leq D_i$ for some $D_* \in (0, \infty)$ and all $1 \leq i \leq M$,
- ② $0 < \phi_* \leq (\partial_T \phi)(T, c, x)$ for some $\phi_* \in (0, \infty)$ and all T, c, x .

Then for every $(\gamma, \tau) \in \left(W^{\frac{1}{2},2}(\partial\Omega)\right)^{M+1}$ there is a weak solution $(c, T) \in (H^1(\Omega))^{M+1}$ of the system of equations

$$\operatorname{div}(D_i(T, c, x) \nabla c_i) = 0, \quad \Delta(\phi(T, c, x)) = q \cdot c \text{ in } \Omega, \quad (c, T)|_{\partial\Omega} = (\gamma, \tau).$$

(1) is physically motivated.

(2) is mathematically motivated $\leftrightarrow$ ellipticity in T .

THE FORWARD PROBLEM: (PARTIAL) UNIQUENESS

$$\boxed{\operatorname{div}(D_i \nabla c_i) = 0, \Delta(\phi(T, c, x)) = q \cdot c \text{ in } \Omega, (c, T)|_{\partial\Omega} = (\gamma, \tau)} \quad (\star)$$

PARTIAL UNIQUENESS

Under same assumptions as before:

- 1 If $(\gamma, \tau) \in \mathbb{R}^M \times W^{\frac{1}{2},2}(\partial\Omega) \subset \left(W^{\frac{1}{2},2}(\partial\Omega)\right)^{M+1}$, then the weak solution $(c, T) \in (H^1(\Omega))^{M+1}$ solving $(\star)$ is unique.
- 2 If $D_i(T, c, x) = D_i(x)$, then for all $(\gamma, \tau) \in \left(W^{\frac{1}{2},2}(\partial\Omega)\right)^{M+1}$ the weak solution of $(\star)$ is unique.

THE INVERSE PROBLEM

- Goal:

- 1 Reconstruct the electric potential $\phi = \phi(T, c, x)$ (main goal)
- 2 Reconstruct the diffusion coefficients $D_i = D_i(T, c, x)$ (secondary goal)

THE INVERSE PROBLEM

- Goal:

- 1 Reconstruct the electric potential $\phi = \phi(T, c, x)$ (main goal)
- 2 Reconstruct the diffusion coefficients $D_i = D_i(T, c, x)$ (secondary goal)

MEASUREMENTS AND INPUT (SIMPLIFIED)

- 1 We can prescribe

$$(c, T)|_{\partial\Omega}$$

- 2 Potential difference on $\partial\Omega$, for all $(\gamma, \tau) \in \left(W^{\frac{1}{2},2}(\partial\Omega)\right)^{M+1}$ we know

$$\phi(\gamma(x), \tau(x), x) - \phi(\gamma(y), \tau(y), y) \text{ for all } y, x \in \partial\Omega;$$

- 3 We measure temperature change and particle flux

$$\mathcal{N} \cdot \nabla T \text{ and } \mathcal{N} \cdot (D_i(T, c, x) \nabla c_i) \text{ for all } 1 \leq i \leq M$$

NECESSITY OF INTERIOR MEASUREMENTS

- If $D_i = D_i(x)$, we have unique solution $\rightarrow$ Dirichlet to Neumann map

$$\Lambda_{\text{DN}}[\phi, D_i](\gamma, \tau) := (\mathcal{N} \cdot (D_1 \nabla c_1), \dots, \mathcal{N} \cdot (D_M \nabla c_M), \mathcal{N} \cdot \nabla T)$$

NECESSITY OF INTERIOR MEASUREMENTS

- If $D_i = D_i(x)$, we have unique solution $\rightarrow$ Dirichlet to Neumann map

$$\Lambda_{\text{DN}}[\phi, D_i](\gamma, \tau) := (\mathcal{N} \cdot (D_1 \nabla c_1), \dots, \mathcal{N} \cdot (D_M \nabla c_M), \mathcal{N} \cdot \nabla T)$$

NON-UNIQUENESS (ALBERTI, G., SANTACESARIA [2026])

Let $\phi, \Phi \in C^1$ be globally Lipschitz and "strongly increasing in T " and $D_i = D_i(x)$. Then following are equivalent

- ① $\phi(\gamma(x), \tau(x), x) - \phi(\gamma(y), \tau(y), y) = \Phi(\gamma(x), \tau(x), x) - \Phi(\gamma(y), \tau(y), y)$ for all γ, τ on $\partial\Omega$
- ② $\Lambda_{\text{DN}}[\phi, D_i] = \Lambda_{\text{DN}}[\Phi, D_i]$
- ① For some $r \in \mathbb{R}$, $\Phi(c, T, x) = \phi(c, T, x) + r$ for all $(c, T, x) \in \mathbb{R}^{M+1} \times \partial\Omega$
- ② $\mathcal{N} \cdot (\nabla_x \phi)(c, T, x) = \mathcal{N} \cdot (\nabla_x \Phi)(c, T, x)$ for all $(c, T, x) \in \mathbb{R}^{M+1} \times \partial\Omega$

A MIXED INVERSE PROBLEM

NEW MEASUREMENTS

- ① We prescribe $(c, T)|_{\partial\Omega}$
- ② We measure " $\phi(x) - \phi(y)$ " on $\partial\Omega$ (boundary electric potential)
- ③ $\mathcal{N} \cdot (D_i(T, c, x) \nabla c_i(x))$ (boundary particle flux)
- ④ **Instead of $\mathcal{N} \cdot \nabla T$, now measure $T(x)$ inside Ω (interior temperature)**

A MIXED INVERSE PROBLEM

NEW MEASUREMENTS

- ① We prescribe $(c, T)|_{\partial\Omega}$
- ② We measure " $\phi(x) - \phi(y)$ " on $\partial\Omega$ (boundary electric potential)
- ③ $\mathcal{N} \cdot (D_i(T, c, x) \nabla c_i(x))$ (boundary particle flux)
- ④ Instead of $\mathcal{N} \cdot \nabla T$, now measure $T(x)$ inside Ω (interior temperature)

MAIN RESULT (INFORMAL)

If two pairs of electric potentials and diffusion coefficients (ϕ, D) , $D = (D_1, \dots, D_M)$, and $(\tilde{\phi}, \tilde{D})$, $\tilde{D} = (\tilde{D}_1, \dots, \tilde{D}_M)$ are compatible with the above measurements, then

$$D = \tilde{D} \text{ and } \tilde{\phi} = \phi + r \text{ for some } r \in \mathbb{R} \text{ everywhere.}$$

Here $\phi = \phi(c, T, x)$, $D = D(c, T, x)$.

PROOF IDEA: BACK TO THE ROOTS

- *Classical Calderón problem:*

$\operatorname{div}(\gamma(x)\nabla u(x)) = 0$, prescribe $u|_{\partial\Omega}$, measure $\mathcal{N} \cdot (\gamma\nabla u)$, find γ

PROOF IDEA: BACK TO THE ROOTS

- *Classical Calderón problem:*

$$\operatorname{div}(\gamma(x)\nabla u(x)) = 0, \text{ prescribe } u|_{\partial\Omega}, \text{ measure } \mathcal{N} \cdot (\gamma\nabla u), \text{ find } \gamma$$

- *Quasi-linear Calderón problem:*

$$\operatorname{div}(\gamma(u, x)\nabla u(x)) = 0, \text{ prescribe } u|_{\partial\Omega}, \text{ measure } \mathcal{N} \cdot (\gamma\nabla u), \text{ find } \gamma$$

PROOF IDEA: BACK TO THE ROOTS

- *Classical Calderón problem:*

$$\operatorname{div}(\gamma(x)\nabla u(x)) = 0, \text{ prescribe } u|_{\partial\Omega}, \text{ measure } \mathcal{N} \cdot (\gamma\nabla u), \text{ find } \gamma$$

- *Quasi-linear Calderón problem:*

$$\operatorname{div}(\gamma(u, x)\nabla u(x)) = 0, \text{ prescribe } u|_{\partial\Omega}, \text{ measure } \mathcal{N} \cdot (\gamma\nabla u), \text{ find } \gamma$$

LINEARISATION LEMMA, SUN [1996]

Fix $z \in \mathbb{R}$ and $f \in C^\infty(\partial\Omega)$. Then

$$\frac{\mathcal{N} \cdot (\gamma(u, x)\nabla u)}{t} \rightarrow \mathcal{N}(x) \cdot (\gamma(z, x)\nabla w) \text{ as } t \rightarrow 0$$

$$\operatorname{div}(\gamma(u, x)\nabla u) = 0, u|_{\partial\Omega} = z + tf \text{ and } \operatorname{div}(\gamma(z, x)\nabla w) = 0, w|_{\partial\Omega} = f.$$

PROOF IDEA: A GENERALISED LINEARISATION LEMMA

LINEARISATION LEMMA

Let $A : (L^2(\Omega))^M \rightarrow (L^2(\Omega))^N$ satisfy

- ① A is globally Lipschitz continuous,
- ② For $\mu \in \mathbb{R}^M \subset (L^2(\Omega))^M$: $A(\mu) \in (C^{0,\beta}(\overline{\Omega}))^N$.

Then for fixed $\mu \in \mathbb{R}^M$, $f \in (C^{1,\beta}(\partial\Omega))^M$ we have

$$t^{-1}\mathcal{N} \cdot (\gamma_i(A(u)(x), x) \nabla u_i) \rightarrow \mathcal{N} \cdot (\gamma_i(A(\mu)(x), x) \nabla w_i) \text{ as } t \rightarrow 0,$$
$$\operatorname{div} (\gamma_i(A(u)(x), x) \nabla u_i) = 0, \quad u|_{\partial\Omega} = \mu + tf, \quad 1 \leq i \leq M, \tag{1}$$

$$\operatorname{div} (\gamma_i(A(\mu)(x), x) \nabla w_i) = 0, \quad w|_{\partial\Omega} = f, \quad 1 \leq i \leq M. \tag{2}$$

- Sun's lemma: $M = 1 = N$, $A = \operatorname{Id}$.
- Linearised equations are decoupled classical Calderón problem's $\square$

